Vector Fields and Line Integrals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Vector fields and divergence**

For the vector field $\mathbf{F}(x, y) = \langle x^2 y,\ xy^2 \rangle$, find the divergence $\nabla \cdot \mathbf{F}$.

$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(xy^2) = 2xy + 2xy = 4xy$.

2 **[Foundational]** • **Target: Line integrals**

Evaluate the line integral $\int_C (x + y)\, ds$ where C is the line segment from $(0, 0)$ to $(3, 4)$.

Parametrize $x = 3t$, $y = 4t$, $0 \le t \le 1$: $ds = 5\, dt$. $\int_0^1 (3t + 4t)(5)\, dt = 35 \int_0^1 t\, dt = \frac{35}{2}$.

3 **[Foundational]** • **Target: Line integrals**

Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y, x \rangle$ along $\mathbf{r}(t) = \langle t, t^2 \rangle$, $0 \le t \le 1$.

$\mathbf{F}(\mathbf{r}(t)) = \langle t^2, t \rangle$, $\mathbf{r}'(t) = \langle 1, 2t \rangle$: $\int_0^1 (t^2 + 2t^2)\, dt = \int_0^1 3t^2\, dt = 1$.

4 **[Foundational]** • **Target: Conservative vector fields and potential functions**

Determine whether $\mathbf{F}(x, y) = \langle 2xy, x^2 + 3y^2 \rangle$ is conservative, using the test $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

$P = 2xy$: $\frac{\partial P}{\partial y} = 2x$. $Q = x^2 + 3y^2$: $\frac{\partial Q}{\partial x} = 2x$. Since these are equal, $\mathbf{F}$ is conservative.

5 **[Foundational]** • **Target: Conservative vector fields and potential functions**

Find a potential function $f(x, y)$ such that $\nabla f = \mathbf{F}$ for the field above.

$f_x = 2xy \Rightarrow f = x^2 y + g(y)$. Then $f_y = x^2 + g'(y) = x^2 + 3y^2$, so $g'(y) = 3y^2$, giving $g(y) = y^3$. Potential function: $f(x, y) = x^2 y + y^3$ (plus any constant).

6 **[Foundational]** • **Target: The Fundamental Theorem for line integrals**

Using the potential function $f(x, y) = x^2 y + y^3$ found above, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0, 0)$ to $(1, 2)$, using the Fundamental Theorem for line integrals.

$\int_C \mathbf{F} \cdot d\mathbf{r} = f(1, 2) - f(0, 0) = (1 \cdot 2 + 8) - (0) = 10$.

7 **[Foundational]** • **Target: Green's Theorem**

Use Green's Theorem, $\oint_C P\, dx + Q\, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$, to evaluate $\oint_C xy\, dx + x^2\, dy$ where C is the boundary of the square $0 \le x \le 1$, $0 \le y \le 1$.

$P = xy$, $Q = x^2$: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 2x - x = x$. $\iint_D x\, dA = \int_0^1 \int_0^1 x\, dy\, dx = \int_0^1 x\, dx = \frac{1}{2}$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Green's Theorem**

Use Green's Theorem to find the area of the region enclosed by a curve C, using the formula $A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$, for the circle $x = 2\cos t$, $y = 2\sin t$, $0 \le t \le 2\pi$.

$dx = -2\sin t\,dt$, $dy = 2\cos t\,dt$. $x\,dy - y\,dx = 2\cos t(2\cos t\,dt) - 2\sin t(-2\sin t\,dt) = 4\cos^2 t\,dt + 4\sin^2 t\,dt = 4\,dt$. $A = \frac{1}{2}\int_0^{2\pi} 4\,dt = 4\pi$ (matching the known area of a radius-2 circle).

9 **[Intermediate]** • **Target: Green's Theorem**

Use Green's Theorem to evaluate $\oint_C (3y)\,dx + (5x)\,dy$ where C is the circle $x^2 + y^2 = 4$, traversed counterclockwise.

$P = 3y$, $Q = 5x$: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 5 - 3 = 2$. $\iint_D 2\,dA = 2 \cdot 4\pi = 8\pi$, using that the disk of radius 2 has area 4π.

10 **[Intermediate]** • **Target: Work done by a force field**

A force field $\mathbf{F}(x, y) = \langle y, x \rangle$ acts on an object moving along the straight path from $(0, 0)$ to $(2, 0)$ then from $(2, 0)$ to $(2, 3)$. Find the total work done, $W = \int_C \mathbf{F} \cdot d\mathbf{r}$, by evaluating each leg separately.

Leg 1 ($y = 0$, $x : 0 \to 2$): $\mathbf{F} = \langle 0, x \rangle$, $d\mathbf{r} = \langle dx, 0 \rangle$: contributes $\int_0^2 0\,dx = 0$. Leg 2 ($x = 2$, $y : 0 \to 3$): $\mathbf{F} = \langle y, 2 \rangle$, $d\mathbf{r} = \langle 0, dy \rangle$: contributes $\int_0^3 2\,dy = 6$. Total work $= 0 + 6 = 6$.

11 **[Intermediate]** • **Target: Work done by a force field**

Confirm the work found above using the potential function $f(x, y) = xy$ (since $\mathbf{F} = \langle y, x \rangle = \nabla f$) and the Fundamental Theorem for line integrals.

$W = f(2, 3) - f(0, 0) = 6 - 0 = 6$, matching the piecewise calculation — confirming $\mathbf{F}$ is indeed conservative with potential $f(x, y) = xy$.

12 **[Intermediate]** • **Target: Testing for conservative fields**

Determine whether $\mathbf{F}(x, y) = \langle y^2, 2xy + 1 \rangle$ is conservative, and if so, find a potential function.

$P = y^2$: $\frac{\partial P}{\partial y} = 2y$. $Q = 2xy + 1$: $\frac{\partial Q}{\partial x} = 2y$. Equal, so conservative. $f_x = y^2 \Rightarrow f = xy^2 + g(y)$;
$f_y = 2xy + g'(y) = 2xy + 1$, so $g'(y) = 1$, $g(y) = y$. Potential function: $f(x, y) = xy^2 + y$.

13 **[Intermediate]** • **Target: Testing for conservative fields**

Determine whether $\mathbf{F}(x, y) = \langle 3x^2y, x^3 + 2y \rangle$ is conservative, and find a potential function if so.

$P = 3x^2y$: $\frac{\partial P}{\partial y} = 3x^2$. $Q = x^3 + 2y$: $\frac{\partial Q}{\partial x} = 3x^2$. Equal, so conservative. $f_x = 3x^2y \Rightarrow f = x^3y + g(y)$;
$f_y = x^3 + g'(y) = x^3 + 2y$, so $g'(y) = 2y$, $g(y) = y^2$. Potential function: $f(x, y) = x^3y + y^2$.

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14 **[Intermediate]** • **Target: Applying Green's Theorem to circulation**

Use Green's Theorem to evaluate the circulation $\oint_C (-y)\,dx + x\,dy$ around the boundary of the rectangle $0 \le x \le 3$, $0 \le y \le 2$, traversed counterclockwise.

$P = -y$, $Q = x$: $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 1 - (-1) = 2$. $\iint_D 2\,dA = 2(3)(2) = 12$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Line integrals over piecewise paths** • **Cross-Lesson: 9.01 Separable Differential Equations**

Evaluate $\int_C y\,dx + x\,dy$ where C consists of the segment from $(0, 0)$ to $(1, 0)$ followed by the segment from $(1, 0)$ to $(1, 1)$.

Leg 1 ($y = 0$, $x : 0 \to 1$): $y\,dx + x\,dy = 0\,dx + x(0) = 0$: contributes 0. Leg 2 ($x = 1$, $y : 0 \to 1$): $y(0) + 1\,dy$: contributes $\int_0^1 dy = 1$. Total $= 0 + 1 = 1$.

16 **[Challenging]** • **Target: Line integrals over piecewise paths** • **Cross-Lesson: 1.01 Precalculus Review for Calculus**

Confirm the result above using the potential function $f(x, y) = xy$ (since $\langle y, x \rangle = \nabla f$) and the endpoints $(0, 0)$ and $(1, 1)$.

$f(1, 1) - f(0, 0) = 1 - 0 = 1$, matching the piecewise calculation.

17 **[Challenging]** • **Target: Flux and Green's Theorem (normal form)** • **Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves**

Using the divergence form of Green's Theorem, $\oint_C \mathbf{F} \cdot \mathbf{n}\,ds = \iint_D \nabla \cdot \mathbf{F}\,dA$, find the outward flux of $\mathbf{F}(x, y) = \langle x, y \rangle$ across the boundary of the disk $x^2 + y^2 \le 4$.

$\nabla \cdot \mathbf{F} = 1 + 1 = 2$. $\iint_D 2\,dA = 2 \cdot (\pi \cdot 2^2) = 8\pi$.

18 **[Challenging]** • **Target: A capstone conservative-field application** • **Cross-Lesson: 11.01 Sequences and Series**

A particle moves through a force field $\mathbf{F}(x, y) = \langle 2xy^3, 3x^2y^2 + 4 \rangle$ from the point $(0, 0)$ to $(2, 1)$ along any path. First verify that $\mathbf{F}$ is conservative, then find a potential function $f(x, y)$, and finally use it to compute the total work done by the field on the particle.

Test: $P = 2xy^3$: $\frac{\partial P}{\partial y} = 6xy^2$. $Q = 3x^2y^2 + 4$: $\frac{\partial Q}{\partial x} = 6xy^2$. Equal, so conservative. Potential: $f_x = 2xy^3 \Rightarrow f = x^2y^3 + g(y)$; $f_y = 3x^2y^2 + g'(y) = 3x^2y^2 + 4$, so $g'(y) = 4$, $g(y) = 4y$: $f(x, y) = x^2y^3 + 4y$. Work $= f(2, 1) - f(0, 0) = (4 + 4) - 0 = 8$.

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19 [Challenging] • Target: A capstone conservative-field application • Cross-Lesson: 12.01 Convergence Tests

For the vector field $\mathbf{F}(x, y, z) = \langle yz, xz, xy + 2z \rangle$, complete each step:

a Find the divergence $\nabla \cdot \mathbf{F}$. $\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(yz) + \frac{\partial}{\partial y}(xz) + \frac{\partial}{\partial z}(xy + 2z) = 0 + 0 + 2 = 2$.

b Find the curl $\nabla \times \mathbf{F}$. $\nabla \times \mathbf{F} = \left\langle \frac{\partial}{\partial y}(xy + 2z) - \frac{\partial}{\partial z}(xz),\ \frac{\partial}{\partial z}(yz) - \frac{\partial}{\partial x}(xy + 2z),\ \frac{\partial}{\partial x}(xz) - \frac{\partial}{\partial y}(yz) \right\rangle = \langle x - x,\ y - y,\ z - z \rangle = \langle 0, 0, 0 \rangle$.

c Based on the curl, determine whether $\mathbf{F}$ could be conservative in 3D. Since $\nabla \times \mathbf{F} = \mathbf{0}$ everywhere, $\mathbf{F}$ is conservative (irrotational) in 3D.

(a) $\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(yz) + \frac{\partial}{\partial y}(xz) + \frac{\partial}{\partial z}(xy + 2z) = 0 + 0 + 2 = 2$.; (b) $\nabla \times \mathbf{F} = \left\langle \frac{\partial}{\partial y}(xy + 2z) - \frac{\partial}{\partial z}(xz),\ \frac{\partial}{\partial z}(yz) - \frac{\partial}{\partial x}(xy + 2z),\ \frac{\partial}{\partial x}(xz) - \frac{\partial}{\partial y}(yz) \right\rangle = \langle x - x,\ y - y,\ z - z \rangle = \langle 0, 0, 0 \rangle$.; (c) Since $\nabla \times \mathbf{F} = \mathbf{0}$ everywhere, $\mathbf{F}$ is.

20 [Challenging] • Target: Line integrals over piecewise paths • Cross-Lesson: 12.02 Power Series and Taylor Series

Level 3 · Advanced Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^3, y^3, z^3 \rangle$ across the boundary sphere $S : x^2 + y^2 + z^2 = 1$.

Compute the divergence: $abla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^3) + \frac{\partial}{\partial y}(y^3) + \frac{\partial}{\partial z}(z^3) = 3x^2 + 3y^2 + 3z^2 = 3(x^2 + y^2 + z^2)$ Apply the Divergence Theorem: $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E 3(x^2 + y^2 + z^2)\, dV$ Convert to spherical coordinates: $x^2 + y^2 + z^2 = \rho^2$, $dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Vector fields and divergence

Divergence theorem: $\oiint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (abla \cdot \mathbf{F})\, dV$. For $\mathbf{F} = \mathbf{r} = (x, y, z)$, $abla \cdot \mathbf{F} = 3$: flux through unit sphere =

a. 3

b. $3V = 4\pi$ ✓

c. 4π

d. 12π

Applying Vectors and Vector-Valued Functions principles, $3 \cdot V = 3 \cdot 4\pi/3 = 4\pi$. Therefore, the verified correct option is $3V = 4\pi$.

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22 [Foundational] • Target: Line integrals

Work done by $\mathbf{F} = (y, x, 0)$ along line from $(0, 0, 0)$ to $(1, 1, 0)$:

a. 0

b. 2

c. 1 ✓

d. 1/2

$\mathbf{F} = abla(xy)$ conservative; $W = xy|_{(0,0)}^{(1,1)} = 1$.

23 [Foundational] • Target: Conservative vector fields and potential functions

The divergence theorem applied to $\mathbf{F} = (x^3, y^3, z^3)$ over unit ball gives $\oiint_S \mathbf{F} \cdot \hat{n}\, dS = \iiint_V 3(x^2 + y^2 + z^2) dV =$

a. $4\pi/3$

b. 3π

c. $12\pi/5$ ✓

d. π

$\int_0^1 3r^2 \cdot 4\pi r^2 dr = 12\pi \int_0^1 r^4 dr = 12\pi/5$.

24 [Foundational] • Target: The Fundamental Theorem for line integrals

Green's theorem in the form $\iint_D (\partial Q/\partial x - \partial P/\partial y) dA = \oint_C P dx + Q dy$. For $P = 0, Q = x$: $\oint_C x\, dy =$

a. $-\iint_D dA$

b. 0

c. $\iint_D dA = \text{Area}(D)$ ✓

d. $\oint_C y\, dx$

$Q_x = 1 \Rightarrow \iint 1\, dA = \text{Area}$.

25 [Foundational] • Target: Green's Theorem

A conservative vector field $\mathbf{F} = ablaf$ has $abla \times \mathbf{F} =$

a. $\mathbf{F}$

b. f

c. $ablaf$

d. $\mathbf{0}$ ✓

Applying Vectors and Vector-Valued Functions principles, Conservative ■ curl-free. Therefore, the verified correct option is **0**.

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26 [Foundational] • Target: Work done by a force field

Line integral of scalar field: $\int_C f\,ds = \int_a^b f(\mathbf{r}(t))|\mathbf{r}'(t)|dt$. For $f = 1$, $\int_C 1\,ds =$

a. area

b. volume

c. 0

d. **arc length ✓**

Applying Vectors and Vector-Valued Functions principles, $\int_C ds = L$ (arc length). Therefore, the verified correct option is arc length.

27 [Foundational] • Target: Testing for conservative fields

Normal line to $z = x^2 + y^2$ at $(1, 1, 2)$: direction:

a. $(2, 2, -1)$ ✓

b. $(1, 1, 1)$

c. $(-2, -2, 1)$

d. $(1, 0, -2)$

Normal to surface $F = z - x^2 - y^2 = 0$: $abla F = (-2x, -2y, 1) = (-2, -2, 1)$. Direction: $(-2, -2, 1)$ or $(2, 2, -1)$.

28 [Foundational] • Target: Applying Green's Theorem to circulation

Find the flux of $\mathbf{F} = \langle x, y, z\rangle$ through the closed unit sphere using the Divergence Theorem.

a. 2π

b. $\frac{4\pi}{3}$

c. 4π ✓

d. $\frac{8\pi}{3}$

$abla \cdot \mathbf{F} = 3$. Volume of unit sphere $= \frac{4\pi}{3}$. Flux $= 3 \cdot \frac{4\pi}{3} = 4\pi$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Line integrals over piecewise paths

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x^2, y^2, z^2\rangle$ across the closed cylinder $x^2 + y^2 \le 4,\ 0 \le z \le 2$.

a. 32π

b. 8π

c. -16π

d. 16π ✓

$abla \cdot \mathbf{F} = 2x + 2y + 2z$. The flux equals $\iiint_E abla \cdot \mathbf{F}\,dV = 16\pi$.

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30 [Intermediate] • Target: Flux and Green's Theorem (normal form)

Use the Divergence Theorem to find the outward flux of $\mathbf{F} = \langle x, y, z \rangle$ across the cube $[0, 1]^3$.

a. 3 ✓

b. 6

c. $\frac{3}{2}$

d. -3

$abla \cdot \mathbf{F} = 3$. The flux equals $\iiint_E abla \cdot \mathbf{F}\, dV = 3$.

31 [Intermediate] • Target: A capstone conservative-field application

Stokes' theorem for $\mathbf{F} = (y, -x, 0)$ over unit disk $z = 0$: $\oint \mathbf{F} \cdot d\mathbf{r} =$

a. 0

b. 2π

c. -2π ✓

d. π

$abla \times \mathbf{F} = (0, 0, -2) \Rightarrow \iint (0, 0, -2) \cdot (0, 0, 1) dA = -2\pi$.

32 [Intermediate] • Target: Vector fields and divergence

Flux integral $\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v)\, dA$. For $\mathbf{F} = (0, 0, 1)$ over upward-facing plane $z = 0$, $D = \{x^2 + y^2 \le 1\}$:

a. 0

b. 1

c. 2π

d. π ✓

$\iint_D (0, 0, 1) \cdot (0, 0, 1) dA = \pi r^2 = \pi$.

33 [Intermediate] • Target: Line integrals

Green's theorem: $\oint_C y^2\, dx + 2xy\, dy$ over boundary of $[0, 2] \times [0, 3]$:

a. 6

b. 0 ✓

c. 12

d. 3

$Q_x - P_y = 2y - 2y = 0 \Rightarrow \iint 0\, dA = 0$.

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34 [Intermediate] • Target: Conservative vector fields and potential functions

Gradient theorem (FTC for line integrals): $\int_C abla f \cdot d\mathbf{r} =$

a. $\iint f\,dA$

b. $f(\mathbf{r}(b)) - f(\mathbf{r}(a))$ ✓

c. $\oint f\,ds$

d. $abla^2 f$

Path-independence for gradient fields.

35 [Intermediate] • Target: The Fundamental Theorem for line integrals

Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle 0, 0, z\rangle$ across the upper hemisphere $x^2 + y^2 + z^2 = 1,\ z \geq 0$, with outward normal.

a. $\frac{4\pi}{3}$

b. $-\frac{2\pi}{3}$

c. $\frac{2\pi}{3}$ ✓

d. $\frac{\pi}{3}$

With $\mathbf{F} \cdot \mathbf{n} = z^2/1$ on the sphere, integrating in spherical coordinates gives $\frac{2\pi}{3}$ (equivalently, half the flux $\frac{4}{3}\pi a^3$ through the whole sphere).

36 [Intermediate] • Target: Green's Theorem

Use Stokes' theorem to find the circulation $\oint_C \mathbf{F} \cdot d\mathbf{r}$ of $\mathbf{F} = \langle y,\ -x, xz\rangle$ around the circle $x^2 + y^2 = 9$ in the plane $z = 0$, oriented counterclockwise seen from above.

a. -18π ✓

b. 18π

c. -9π

d. -36π

$abla \times \mathbf{F} = \langle 0,\ -z,\ -2\rangle$. Taking the flat surface inside C with $\mathbf{n} = \mathbf{k}$, $\iint_S (abla \times \mathbf{F}) \cdot \mathbf{k}\, dS = -18\pi$.

37 [Intermediate] • Target: Work done by a force field

Evaluate $\oint_C (y\,dx - x\,dy)$ around the circle $x^2 + y^2 = 4$ (counterclockwise) using Green's Theorem.

a. 4π

b. 8π

c. -4π

d. -8π ✓

Green's Theorem: $\iint_D \left(\frac{\partial(-x)}{\partial x} - \frac{\partial y}{\partial y}\right) dA = \iint_D (-1-1)dA = -2 \cdot \pi(4) = -8\pi$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Testing for conservative fields • Cross-Lesson: 11.01 Sequences and Series

Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle 1, 1, 1 \rangle$ across the part of the plane $x + y + z = 3$ in the first octant, with upward normal.

a. $\frac{27}{2}$ ✓

b. 27

c. $-\frac{27}{2}$

d. $\frac{27}{4}$

Write $z = -x - y + 3$; $d\mathbf{S} = \langle 1, 1, 1 \rangle \, dA$. The flux is $\iint_D \mathbf{F} \cdot \langle 1, 1, 1 \rangle \, dA = \frac{27}{2}$.

39 [Challenging] • Target: Applying Green's Theorem to circulation • Cross-Lesson: 12.01 Convergence Tests

Mean Value Theorem for Integrals: if f is continuous on $[a, b]$, there exists $c \in (a, b)$ with $\int_a^b f(x)\,dx =$

a. $f(c)(b - a)$ ✓

b. $f'(c)(b - a)$

c. $f(b) - f(a)$

d. $f(c)$

This is the integral form of the MVT, closely tied to FTC Part 1: the average value $f(c)$ times the interval length equals the integral.

40 [Challenging] • Target: Line integrals over piecewise paths • Cross-Lesson: 12.02 Power Series and Taylor Series

For $F(x) = \int_a^x f(t)dt$ with f continuous, the proof of FTC Part 1 uses the Mean Value Theorem for Integrals to write $F(x + h) - F(x)$ as:

a. hf'(x)

b. $f(c) \cdot h$ for some $c \in [x, x + h]$ ✓

c. $f(x) \cdot (x + h)$

d. f(x+h)-f(x)

MVT for integrals gives $\int_x^{x+h} f(t)dt = f(c)h$ for some c between x and $x + h$; letting $h \to 0$, $c \to x$.

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41 [Challenging] • Target: Flux and Green's Theorem (normal form) • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

Where does the line $\mathbf{r}(t) = \langle 3, -3, -1\rangle + t\langle 0, 0, 1\rangle$ meet the plane $3x + y + 3z = 4$?

a. $\left(3, -3, \frac{1}{3}\right)$

b. $\left(3, -3, -\frac{2}{3}\right)$ ✓

c. $\left(3, -3, -\frac{4}{3}\right)$

d. $(3, -3, -1)$

Substitute the line into the plane: $3t + 3 = 4$, so $t = \frac{1}{3}$ and the point is $\left(3, -3, -\frac{2}{3}\right)$.

42 [Challenging] • Target: A capstone conservative-field application • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

Line integral: $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt$. For $\mathbf{F} = \mathbf{r}$, C line from 0 to $(1, 0, 0)$:

a. 0

b. 1

c. 1/2 ✓

d. 2

Applying Vectors and Vector-Valued Functions principles, $\int_0^1 t\, dt = 1/2$. Therefore, the verified correct option is $1/2$.

43 [Challenging] • Target: Vector fields and divergence • Cross-Lesson: 15.01 Double Integrals

Fubini's theorem allows interchanging iterated integrals when:

a. both integrals converge

b. f is continuous

c. $\iint |f| < \infty$ ✓

d. $f \geq 0$

Applying Fundamental Theorem principles, Absolute integrability. Therefore, the verified correct option is $\iint |f| < \infty$.

44 [Challenging] • Target: Line integrals • Cross-Lesson: 2.01 Evaluating Limits

Which theorem converts a surface integral of $abla \times \mathbf{F}$ to a line integral?

a. Green's Theorem

b. **Stokes' Theorem** ✓

c. Gauss's Divergence Theorem

d. Cauchy's Integral Theorem

Stokes' Theorem states $\iint_S (abla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_C \mathbf{F} \cdot d\mathbf{r}$.

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45 [Challenging] • Target: Conservative vector fields and potential functions • Cross-Lesson: 2.02 Continuity

Use Green's theorem to evaluate $\oint_C x - y\,dx + x + y\,dy$, where C is the boundary of the rectangle $[0, 1] \times [0, 3]$, oriented counterclockwise.

a. 6 ✓

b. -6

c. 12

d. 3

$\oint_C P\,dx + Q\,dy = \iint_D (Q_x - P_y)\,dA$ with $Q_x - P_y = 2$. $\int_0^3 \int_0^1 2\,dx\,dy = 6$.

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